SULIT

Faculty of Computing

UNIVERSITI TEKNOLOGI MALAYSIA

FINAL EXAMINATION SEMESTER I , 2017 / 2018

COURSE CODE : SCSJ3323

COURSE NAME : SOFTWARE DESIGN & ARCHITECTURE

SECTION : 3 SCSJ

TIME : 2 HOURS 30 MINUTES

DATE/DAY :

VENUE :

INSTRUCTIONS:

ANSWER ALL QUESTIONS. PLEASE WRITE YOUR ANSWERS IN THE ANSWER BOOKLET.

(Please Write Your Lecture Name and Section in Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This questions paper consists of **six (6)** printed pages including this page.

SECTION A - STRUCTURED QUESTIONS [50 MARKS]

Answer ALL of the following questions.

Question 1 [10 marks]

a. What is the definition of detailed design? Give a difference between architectural design and detailed design. [3 marks]

b. Briefly describe the main purpose of applying the Interface Segregation Principle (ISP) in detailed design? [2 marks]

c. Give an example of ISP implementation using an appropriate class diagram. Explain how ISP is implemented in your example of ISP implementation. [5 marks]

Question 2 [10 marks]

a. Briefly describe the main difference between Architectural Patterns and Design Patterns. [2 marks].

b. Briefly describe the purpose of any four(4) of the following design patterns. [4 marks].

i. Builder
ii. Prototype
iii. Observer
iv. Adaptor
v. Facade
vi. Singleton

c. Identify a suitable design pattern for each description below. [4 marks]

Description	Design Pattern
i. Which design pattern would you use in a situation where if one object changes state, all its dependents are notified and updated automatically, using a one-to-many relationship.	i. _____
ii. Which design pattern would you use to allow designers to compose (large) tree-like designs structures.	ii. _____
iii. Which design pattern would you use to convert the interface of a class into another interface clients expect?	iii. _____
iv. Which design pattern would you use to provide a unified interface to a set of interfaces in a subsystem	iv. _____

Question 3 [10 marks]

a. The class diagram in Figure 1 shows the structure of a design pattern. Name the design pattern that is represented by that class diagram. Describe the roles of each class in that class diagram. [5 marks]

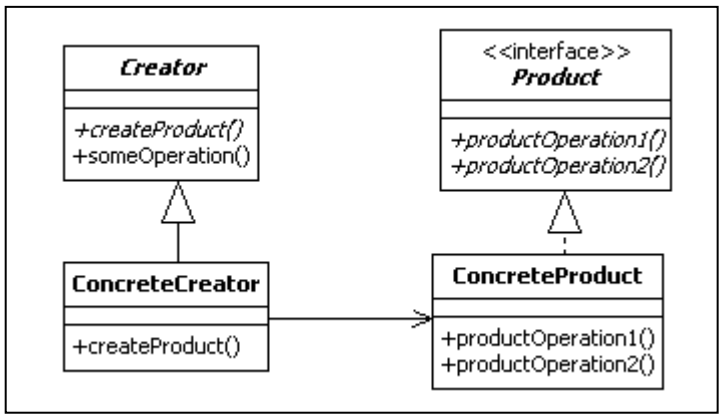


Figure 1

b. The diagram in Figure 2 shows the complex relationships or communications (in terms of function calls) between two subsystems. Which design pattern is suitable to resolve the complexity? Redraw the diagram that applies your selected pattern, and describe two justifications to support your design pattern selection. [5 marks]

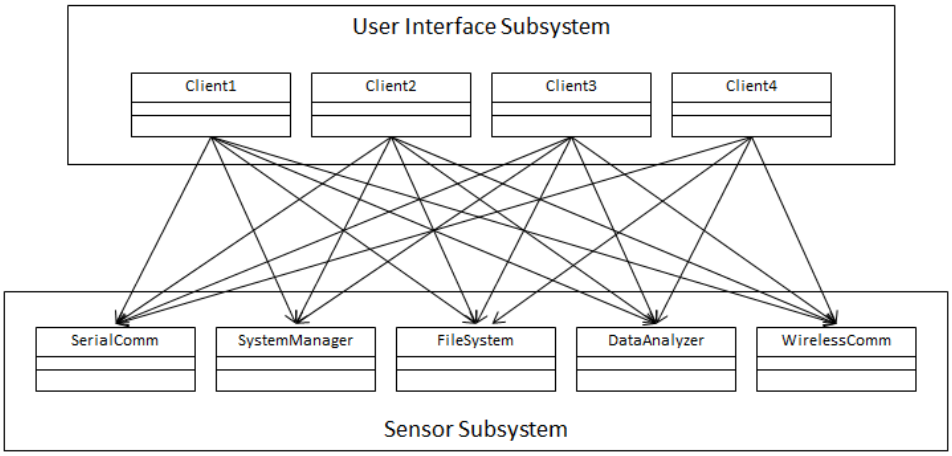


Figure 2

Question 4 [10 marks]

a. What is construction Design? [2 marks]

b. What is the difference between the algorithmic viewpoint and stylistic viewpoint in construction design? Give an example of a technique that can be used for each algorithmic viewpoint and stylistic viewpoint. [5 marks]

c. Name THREE (3) techniques that can be used to evaluate completeness and correctness during construction design. [3 marks]

Question 5 **[10 marks]**

a. Define the TWO (2) main reasons why Software Design Management is important. [2 marks]

b. Discuss the difference between Work Breakdown Structure (WBS) and Linear Responsibility Chart (LRC)? Describe a simple example of LRC that consists of 3 tasks and 2 staffs. [3 marks]

c. Various principles of ethics collectively ensure that professionals in the software engineering field adhere to high levels of ethical conduct. Identify the suitable name of each software engineering principle of ethics according to the provided description. [5 marks]

SE Principle of Ethic	**Description**
i. _____	Software engineers shall act consistently with the public interest.
ii. _____	Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.
iii. _____	Software engineers shall maintain integrity and independence in their professional judgment.
iv. _____	Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
v. _____	Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.

SECTION B - PROBLEM SOLVING [50 MARKS]

Answer ALL of the following questions.

Question 1 [25 marks]

a. Draw a class diagram of the Design Pattern you adopted in your class project. Highlight which Design Pattern you used in that class diagram, give two justifications on why the design pattern is selected, and describe two benefits of using that design pattern. [15 marks]

b. Apply Adapter Design Pattern to the following design problem: [10 marks]

You have developed a Flying Car game, called CAR GAME. In your game, ***Car*** class can do the following functions.

- start()
- forward()
- reverse()
- fly()
- land()
- stop()

You want to improve the Car class so that the forward(), reverse(), fly(), and land() operations may perform faster. But, you think it is a time taking task to modify your code. So, you manage to find code from a friend who has already developed FAST features for his FAST CAR GAME, and he allows you to use his class diagram and code. His ***FastCar*** class consist of the following operations: fastForward(), fastReverse(), fastFly(), and fastLand(). Thus, you want to re-use his class diagram and code to improve your CAR GAME. For this purpose, it is appropriate to apply Adapter Design Pattern.

Your Task: Draw **a class diagram** so that CAR GAME can use the use the functions of FAST CAR GAME. Highlight the adapter class in your diagram.

Question 2 [25 marks]

a) Based on the described Public Library System (PLS), draw a UML Diagram that represents:

i. the state-based model for a **book** object (based on the lifecycle of the book in PLS). [10 marks]

ii. the flow-based model (the algorithmic viewpoint) for the *renewLoan*() function. [7 marks]

Public Library System (PLS)

Public Library System (PLS) is a software system that supports the book cataloguing and borrow management for a community-based public library. Each time a new book is received, the book is registered and given a unique call number. Library users may use PLS to search for a book. If the book is available, users may get it from the shelf based on the call number. To borrow the book, users must bring the book to the counter to process the borrow record. However, if the book is currently borrowed by other library users, the system will give an option to place a booking for the book. Apart from searching and booking, PLS also allows users to renew the book using their web-based system. Users must select the book to be renewed. If the status of the book is 'booked', the current borrower shouldn't be allowed to renew his/her loan and he/she must return the book to the library. In addition, library users are not allowed to renew the book if the book is 'overdue' although the book is not booked by other library users.

b) The project progress for the current month (i.e. June) is shown in the graph below. The total duration of this project is 12 months (Jan-Dec) with the total planned cost is RM120,000.00. Based on the information from the graph, calculate the Cost Performance Index (CPI), Schedule Performance Index (SPI), Estimated Remaining Cost (ETC), and Estimated Total Cost (EAC) at the current month (i.e. June) of this software project. Show the steps of each calculation. [8 marks]

